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RESEARCH ARTICLE



Designing regulatory sandboxes: a comprehensive framework for aligning functionalities and objectives

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ABSTRACT

Regulatory sandboxes have been introduced in jurisdictions worldwide, providing a controlled testing environment for piloting new technologies such as autonomous vehicles and FinTech. We present a general framework for designing a regulatory sandbox that policymakers can operationalize. Within the general framework, we introduce three types of sandbox functionalities: exploratory, confirmatory, and on-demand that can be combined to achieve a range of objectives. The characteristics of each functionality type are presented as distinct design choices, enabling policymakers to design sandboxes along multiple dimensions. The framework emphasizes a deliberative approach for determining whether to introduce a regulatory sandbox, taking into account several factors, including the rationale for undertaking a legal experiment, the existence of experimental legislation, and the limits imposed by current legislation. This article contributes to the literature by providing a comprehensive framework for designing regulatory sandboxes.

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1. Introduction

Regulatory sandboxes are experimental legal regimes which allow regulatees to conduct experiments involving innovative products or services. Existing laws and regulations often prevent firms from engaging in experiments. Hence, to encourage innovation and allow for such experiments, regulators introduce sandboxes whereby firms can propose projects and experiments and apply to enter sandboxes. Regulators will award legal derogations to successful applicants, which allow firms to conduct limited experiments in real-world settings.

Sandboxes were first used by the UK's Financial Conduct Authority ("FCA") in 2016 as a tool to increase innovation in the financial technology ("FinTech") sector. Since then, regulatory sandboxes have been deployed in other jurisdictions such as Australia, Singapore, the Netherlands, and some US states (Gerlach and Rugilo 2019). Sandboxes have also been deployed in other sectors and applications, such as the energy sector, robotics in healthcare in Singapore, and autonomous vehicles ("AVs") (Schittekatte, Reif, and Meeus 2021; Tan and Taeiagh 2021a, 2021b; van der Waal, Das, and van der Schoor 2020).

Despite the rising popularity of sandboxes, several fundamental issues remain unaddressed. First, although multiple sandbox designs have been implemented across various sectors, the underlying sandbox design itself has not been thoroughly examined and warrants further investigation. For example, a technical guide by Jeník and Duff (2020) proposes a design process for sandboxes for the financial sector. Although the guide discusses design choices of sandboxes, several factors are overlooked or not examined substantially, such as monitoring processes within sandboxes. The guide proposes a diligence framework for planning which focuses on contextual factors such as jurisdictional, institutional, and market characteristics, it does not explore applicability of sandboxes in other use cases and sectors. Other attempts to address sandbox design have been highly context-specific and tailored to particular sectors and jurisdictions, such as Allen (2019) which focuses solely on entry

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and authorization of participants in US FinTech sandboxes. Second, inadequate focus on sandbox design has resulted in undesirable outcomes. A study of sandboxes within the European Union's (EU) energy sector found considerable design variations, in particular different approaches to regulators' involvement. Several sandboxes were designed and implemented without involving relevant ministries, which could limit regulatory learning and the scope of the sandbox (Beckstedde et al. 2023). Third, sandbox functionalities are often overlooked in sandbox design. We define sandbox functionality as an assemblage of features or operations which enables the sandbox to achieve its specific objectives. There has been limited discussion distinguishing between various functionalities as well as the different types and purposes of experiments related to functionalities.

The key research question of this article is "How can regulatory sandboxes be designed with different functionalities and for various use cases and sectors?" We propose a general framework for designing regulatory sandboxes, considering functionalities of sandboxes and design choices based on common sandbox characteristics. Section 2 provides a brief background of regulatory sandboxes. In Section 3, we present a decision flow process for introducing sandboxes based on key considerations of regulators and propose three different functionalities of sandboxes. Section 4 explores different sandbox characteristics, the implications of each design choice, and the relationship between functionalities and characteristics. Section 5 discusses the framework, and Section 6 concludes.

2. Background of regulatory sandboxes

In 2016, six jurisdictions, namely the UK, Hong Kong, Singapore, Malaysia, Abu Dhabi, and Australia, introduced regulatory sandboxes for FinTech (Zetzsche et al. 2017). These initiatives emerged after the global financial crisis, which shifted regulators' priorities from a laissez-faire approach to a stricter regulatory framework (Baker 2013; Zetzsche et al. 2017). FinTech firms, which aimed to integrate new technologies into the financial sector, posed challenges to regulators who needed to balance their objectives of market stability and consumer protection with the need to encourage growth and innovation. Hence, regulatory sandboxes addressed this issue by enabling structured experimentation under time-limited legal derogations.

Sandboxes were initially introduced as restricted or structured experimentation, as participating firms were required to operate within defined boundaries. Although common design features exist, such as eligibility criteria for entry, sandbox designs vary significantly (Gumbo and Chude-Okonkwo 2025). For example, some sandboxes assign case officers to applicants to verify eligibility, provide support during the application process, and evaluate and grant exemption requests (Alaassar, Mention, and Aas 2021). In contrast, Hong Kong's sandbox requires firms to file applications directly, while Australia's sandbox offers automatic approval if applicants meet eligibility criteria (Alaassar, Mention, and Aas 2021). Other design variations include duration of legal derogations, the extent of regulators' engagement with the FinTech ecosystem, and the provision of testing facilities (see Table A1 in the Appendix A for summary of sandboxes and their primary functionality).

3. Designing sandboxes

3.1. Decision flow process for introducing sandboxes

The flexibility of regulatory sandbox design has facilitated its adoption across sectors. However, a more deliberate and structured design approach can enhance benefits for regulators and firms while mitigating potential risks associated with experiments. This section proposes a sequential and deliberative decision-making framework for introducing sandboxes, based on their attributes as legal experiments (see Figure 1; (Makarov and Davydova 2021).

First, regulators should assess whether there are grounds for conducting a legal experiment, as firms may need legal derogations to conduct innovative experiments or pilot projects. In many jurisdictions, existing laws may restrict firms from engaging in pilot projects or experiments or create legal ambiguities that deter experimentation. Hence, regulators must grant legal derogations to provide firms with legal certainty to proceed. However, some jurisdictions or sectors may not require



Figure 1. Decision-making process for regulators when deliberating on introducing regulatory sandboxes.

derogations, as existing regulations already allow firms to conduct experiments. For example, China's FinTech sector has mechanisms akin to a regulatory sandbox, such as Pilot Free Trade Zones (PFTZs) which have been in place since 2013 and offer controlled environments for FinTech experimentation (Liao 2020). Furthermore, statutory laws governing these zones have been amended, which allow a broader scope of experiments than the UK FCA's sandbox, and they also enable FinTech experiments outside the territorial limits of PFTZs when necessary.

After establishing valid grounds for a legal experiment, regulators must determine whether experimental legislation is available. Experimental legislation refers to “laws or more commonly, regulations (secondary legislation) which introduce rules in deviation of existing laws for a fixed period, for a limited group of citizens or territory and which are subject to a periodic or final evaluation” (Ranchordas 2015, p. 33). Such legislation provides the legal foundation for granting derogations that enable firms to conduct innovative experiments. Legal definitions of sandboxes can also be introduced to provide legal certainty (Tomčiak and Škrabka 2024). To ensure consumer safety, the legislation must clearly define the scope, which includes geographical boundaries, objectives, duration, and limits of the derogations. Where greater flexibility is preferred, legislation can be introduced that grants regulatory bodies the authority to design and implement sandboxes. In some cases, amending legislation may not be needed. For example, Latvian financial regulators implemented a sandbox by issuing internal regulations, as the sandbox fell under the complete responsibility of regulators (Raudla et al. 2025).

Third, regulators must ensure that experimental legislation adequately controls and minimizes risks to safety. If such legislation already exists, regulators must examine it to confirm it sufficiently limits the scope of experimentation. For instance, regulators should evaluate whether legislation sufficiently restricts the use of personal data. Data-intensive innovations, such as analytics involving consumer

data, pose inherent security risks and are susceptible to data breaches. To address these concerns, some jurisdictions have enacted sector-specific data protection legislation, while others, like Singapore, have introduced comprehensive data privacy legislation applicable across all sectors (Tan and Taeihagh 2021a). Whereas Hungary's MNB FinTech Regulatory Sandbox has structured consumer protection and data governance requirements within the sandbox framework (Kálmán 2025).

Additionally, experimental legislation should provide mechanisms for consumer redress and compensation for any harm incurred during experiments. Hence, legal derogations should ensure participating firms are liable and accountable for damages from experiments.

The three considerations outlined above focus on ensuring adequate legal safeguards exist for consumers and the public. As technologies tested within sandboxes will evolve and mature, periodic reviews of existing regulations are essential to ensure appropriate adjustments to the limitations or boundary conditions of experimental legislation. Through monitoring legal experiments, regulators can deepen their understanding of emerging technologies, enabling the development of new legislation to govern technological advancements. Such legislation may eventually provide firms with legal certainty required for pilot projects, which eliminates the need for legal experiments. Experimental legislation can also be expanded to accommodate new tests and technologies as regulators and firms gain greater understanding of associated risks and applications.

Once a decision to introduce regulatory sandboxes is made, and adequate legal safeguards and experimental legislation are in place, regulators must determine the sandbox design by selecting functionalities which are aligned with their objectives and goals. This section presents three sandbox functionalities: exploratory, confirmatory, and on-demand.

3.2. Sandbox functionalities

Sandboxes can be designed with single or multiple functionalities to achieve one or more goals.

This section introduces three distinct sandbox functionalities: exploratory, confirmatory, and on-demand, which are conceptualized based on goals and objectives of regulators (see Table 1 for summary of functionalities). Each functionality addresses specific regulatory objectives. The framework enables regulators to align sandbox purposes with appropriate design choices while allowing for the integration of additional features to achieve desired objectives.

The first functionality is exploratory, which has the objective of exploring and testing new innovations. For example, participants in Singapore's Energy Market Authority ("EMA") sandbox can test innovative energy solutions and services. Sandbox applicants have a broad degree of flexibility and can adjust their initial application for resubmission after discussions with EMA (EMA 2019).

Confirmatory functionality focuses on affirming and substantiating uses of technology which have already been explored. This functionality is necessary in scenarios where regulators require evidence that a product or service functions as claimed before it is deployed in real-world settings. For example, Mexican legislation requires firms to demonstrate test-ready business models in the FinTech sandbox before applying for regulatory approval (Larios-Hernandez, 2025). Confirmatory functionality also ensures that existing regulations are appropriately calibrated to maintain consumer safety. For example, Singapore's AV sandbox utilizes confirmatory functionality through a tiered testing

Table 1. Sandbox functionalities, purpose, differences, and similarities.

Functionality	Purpose	Differences	Similarity
Exploratory	Testing new use-cases of technology	- Greater degree of flexibility for entry and authorization - Broad scope for experiments	- Limited duration - Frequent and intense monitoring
Confirmatory	- Affirm and substantiate use of technology - Provide evidence that product or service works	- Entry and authorization for tests to pass pre-defined benchmarks - Scope restricted to confirmatory testing	- Limited duration - Frequent and intense monitoring
On-demand	Focus on applying technology to specific use-cases	- Entry and authorization for pre-defined use cases or technology for experimentation - Scope is restricted to use-case or specific sectors of interest to regulator - Incentives provided for firms to participate	- Limited duration - Frequent and intense monitoring

framework. Participants are subjected to three tiers of tests in increasing order of complexity. The first tier involves circuit-based tests conducted within a private, enclosed area which eliminates risks to public road users. Successful AVs then progress to the second tier, which includes physical road tests and a circuit-based test (Tan and Taeihagh 2021a). A key distinction between confirmatory and exploratory functionalities is that confirmatory functionality involves predefined standards against which firms are evaluated. This enables regulators to assess whether the product or service can be deployed in environments with greater risk.

The objective of on-demand functionality is to foster innovation and experimentation in targeted sectors or specific use cases required by regulators. Regulators may require innovations in targeted sectors or specific use cases and will thus introduce sandboxes. However, firms may not conduct experiments due to high costs, or uncertainty over post-experiment deployment due to lack of regulatory clarity. This can be mitigated through on-demand functionality, where regulators offer financial incentives like grants or subsidies to reduce costs. These incentives can encourage firms to conduct experiments aligned with regulators' goals and interests.

For example, Singapore's Infocomm Media Development Authority ("IMDA") has launched a Privacy Enhancing Technologies ("PETs") Sandbox, limiting activities to pilot projects for solutions which are "mature, in market and available for use" (IMDA 2022, p. 5). Grant support is provided to firms implementing the projects. Similarly, Singapore EMA's Thematic Sandbox is also designed to serve the regulator's strategic interests, through scoped themes or problem statements which indicate the regulator's interests in pilot projects to address specific issues (EMA 2019).

The key difference between on-demand and exploratory functionalities is the focus of experiments. Exploratory functionality tests new use cases for technologies, whereas on-demand functionality emphasizes applying technology to use cases for public deployment after exiting the sandbox. For example, IMDA's PET Sandbox exhibits on-demand functionality, explicitly stating that a "research oriented proposal may not be suitable" for the sandbox (IMDA 2022, p. 5).

These three functionalities are not mutually exclusive. A sandbox can combine multiple functionalities to achieve diverse objectives. For example, a sandbox can integrate on-demand and confirmatory functionalities by specifying themes for innovative technologies, while conducting tests to ensure safety and reliability. Another example is a sandbox combining exploratory and confirmatory functionalities through a tiered approach, where each phase has distinct objectives and rules, which allows for both broad experimentation and rigorous validation.

For example, Singapore's AV sandbox has three tiers. The first two tiers focus on exploratory functions, while the third emphasizes confirmatory functions. Entry into the first tier requires a test assessing the safety driver's ability to intervene during emergencies or AV system malfunctions (Tan and Taeihagh 2021a). Within this tier, firms are permitted to conduct exploratory testing on enclosed circuits. To advance to the second tier, firms must pass additional tests evaluating the AV's robustness, before being allowed to conduct exploratory tests in controlled testbeds. The third tier prioritizes confirmatory functionality. Firms must pass rigorous tests demonstrating their AV's capability to operate safely on public roads, which present more complex conditions and have a higher risk compared to the first two tiers.

4. Characteristics, design choices, and functionality

Regulatory sandboxes share common characteristics, which are the basis for design choices that regulators face (see Table 2 for summary of characteristics; Chen and Taeihagh, *forthcoming*). Sandbox functionalities are the result of the different configurations and design choices taken for each characteristic (see Table 3).

4.1. Duration

Duration specifies the time firms may participate in the sandbox. Regulators assign a pre-determined time limit on legal derogations for firms to conduct experiments. Upon completion of testing or expiration of derogations, firms typically exit the sandbox.

Table 2. Summary of sandbox characteristics and brief descriptions.

Characteristic	Description
Duration	- Length of time firms are allowed to participate in sandbox - Pre-determined limit on legal derogations - Fixed time limit applied to all sandbox participants or a case-by-case basis - Extensions on derogations can be granted, subject to regulators' approval
Entry and Authorization	- Eligibility requirements to participate - Activity-specific. e.g. Type of activity falls under purview of regulator, innovation provides public benefit - Firm-specific. e.g. Head office within jurisdiction, level of staff presence within jurisdiction
Scope	- Specificity for types of experiments and activities. e.g. sector-specific to FinTech - Limitations on physical or virtual boundaries of the sandbox
Monitoring	- Ensure compliance of participating firms - Opportunity for regulators to understand technology and its effects - Frequency of interactions and intensity of monitoring can be dynamic, varying over the lifetime of sandbox participation
Exit Rules	- Pre-determined rules, boundaries, or conditions which provides basis for termination of experiment - Firms that violate rules face expulsion from sandbox and other penalties - Exit rules should consider consumer welfare and risks that experiments pose to public
Transition Plans	- Regulators evaluate experiment outcome and determine subsequent opportunities and pathways for the product or service tested by firms - Go-to-market: experiment passes safety standards, is reliable, provides public benefit, ready for real-world deployment - Reject: experiment is unsuitable to be deployed due to concerns over safety, reliability, or suitability - Amend: experiment can be deployed if amendments are made and experiment undergoes confirmatory testing - Coordination required when multiple regulators are involved in transition plans

Table 3. Sandbox functionalities and characteristics.

	Purpose	Duration	Entry and Authorization	Scope	Monitoring	Exit Rules	Transition plans
Exploratory	Testing new use cases of technologies	Limited, longer term	Greater degree of flexibility	Broad scope Many kinds of experiments and pilot projects General purpose technology	Regulatory learning High frequency and intensity of reporting	Moderate	
Confirmatory	Affirm and substantiate uses of a technology which has already been explored	Limited, short term	Demonstrating and passing pre-defined benchmarks	Innovation has to pass pre-defined benchmark	Pre-defined standards for firms to be evaluated against High frequency and intensity of reporting	Strict	Success in testing leads to go-to-market decision
On-demand	Applying technology to use cases with the intention of making it available to general public upon exiting sandbox	Limited, can be short or long term	Pre-defined use cases or technology for experimentation	Pre-defined by regulator Restricted to use case or sector of interest to regulator	Regulatory learning High frequency of reporting High frequency and intensity of monitoring	Moderate	

Sandbox durations are often rule-based. For example, the UK's FinTech sandbox provides six months, while Malaysia and Thailand allow twelve months (Goo and Heo 2020). Conversely, regulators in Singapore and India determine duration on a case-by-case basis (Goo and Heo 2020). Applicants propose a time period for their experiments, and regulators evaluate the proposals to determine the appropriate duration for the derogation. Extensions are also available, subject to regulatory approval.

4.2. Entry and authorization

Sandbox entry can be cohort-based, like the UK's FinTech sandbox, or a continuous, case-by-case admission framework, as in Singapore's and Hungary's FinTech sandboxes (Kálmán 2025).

Sandboxes impose eligibility criteria for participation. For example, the UK's FCA sandbox uses five criteria, which include innovation, consumer benefit, and readiness for testing (Tsai, Lin, and

Liu 2020). Firm-specific criteria can also be used, which focus on the characteristics of the firm instead of the proposed experiment. For example, the UK FCA requires sandbox firms to have their head offices in the UK and maintain a specific level of staff presence. Another commonly used entry criterion is that the innovation should provide public benefit, as seen in sandboxes in the UK and Singapore (Gerlach and Rugilo 2019).

Authorization processes can involve multiple regulators instead of a single agency. For example, South Korea assigns different ministries for judging sandbox operations and jurisdictional regulations. In cases involving technology outside FinTech, one ministry will judge the proposed experiment, while the other ministry focuses on consumer protection and its impact on other businesses (Lee and Seo 2022). Additionally, multi-sector experiments, such as those involving distributed ledger technology (“DLT”) in the healthcare sector will require approval from regulators with oversight of technology and healthcare. Another multi-regulator model involves all regulators in a single industry forming a joint committee, which enables coordination between different regulators and provides a one-stop shop for sandbox applicants (Gumbo and Chude-Onkonkwo 2025; Plato-Shinar 2021).

Multi-regulator sandboxes can result in regulatory delays if regulators have different goals and compete to advance their own agendas. While one agency may approve a sandbox applicant to promote innovation, another may delay approval and market entry to align with the interests of existing businesses to maintain the status quo (Lee and Seo 2022).

Multiple sources of authority can complicate transition plans when firms are seeking approval for market entry. While the sandbox regulator may be the sole authority during entry and authorization, other regulatory bodies may be required for go-to-market approval after testing. However, since these bodies were not involved in sandbox activities, they may lack sufficient understanding of the innovation, potentially hindering approval. Moreover, regulators may require additional confirmatory tests or impose stricter standards before awarding a license.

This issue is more particularly relevant for general-purpose technologies, such as DLT, which may require more complex entry and authorization arrangements. For example, a firm conducting a pilot experiment using DLT in healthcare would need approval from both technology and healthcare regulators. While the technology regulator may solely oversee entry and authorization into the sandbox, successful pilot experiments would require approval from both regulators for real-world market entry. However, healthcare regulators may lack a comprehensive understanding of the technology and may request additional trials to facilitate regulatory learning. They may also deem the product unsuitable for real-world use, requiring amendments or rejecting it entirely. These possible outcomes create significant uncertainty for firms and reduce the effectiveness of regulatory sandboxes in fostering innovations applicable to the real world.

This issue can be addressed by involving all relevant regulatory bodies in the entry and authorization stage. Regulators responsible for the transition plan and market entry at the end of the sandbox duration should align with those involved in the initial entry and authorization process. In the case of general-purpose technologies, the technology regulator and sector-specific regulators should collaborate during entry and authorization to establish conditions for sandbox admission and pilot experiment limits. The sector regulator should evaluate the proposal and set pre-determined sector-specific conditions for firm entry. If firms propose experiments across multiple sectors and meet conditions in some but not all, they should still be permitted to conduct pilot experiments in sectors where they meet the required conditions.

An example of a multi-regulator sandbox is the European Blockchain Sandbox for Distributed Ledger Technologies. Operated by Bird & Bird, an international law firm appointed by the European Commission, the DLT sandbox involves applicants submitting pilot project proposals for evaluation by a selection team of blockchain experts. The team, designated by Bird & Bird, selects use cases for approval by the European Commission. Bird & Bird then identifies relevant regulators for each use case, with an average of 1.5 regulators per case (European Commission 2023). This structure presents challenges, as the selection of use case owners (firms) is determined by a selection team, instead of regulators, who set market entry conditions for firms.

4.3. Scope

Scope refers to the specificity of experiments and activities allowed within a sandbox, and the boundaries placed on its physical and virtual spaces.

Some sandboxes are sector-specific, limiting experiments to the designated sector. For example, the Netherlands has a regulatory sandbox for the energy sector, allowing experiments involving peer-to-peer energy trading and smart microgrid market structures (Correa, Gomez, and Cossent 2021). Other sandbox designs use exclusion rules to narrow the scope. While most FinTech sandboxes permit cryptoasset activities, India's FinTech sandbox excludes activities like Initial Coin Offerings (ICOs) and cryptocurrency services (Ahern 2021). Additionally, regulators may limit the scope to a specific market segment, such as Hong Kong's InsurTech sandbox, which focuses on insurance technology experiments (Ahern 2021).

Sandbox scope also encompasses the spatial location, whether physical, virtual, or both. Some experiments, such as AVs, require physical environments. Others, like digital technology experiments, may be conducted entirely in virtual settings. Some experiments may require both physical and virtual environments, like payment infrastructure experiments which require a physical payment device and the underlying digital technology.

Another dimension of scope is geographical. Sandboxes can span entire regions, such as the EU, be confined to a country, or be limited to a locality such as a business district within a city. These limits can either be specified by the sandbox structure, or on a case-by-case basis.

The UK FCA's FinTech sandbox is an example of a virtual sandbox confined to the UK's geographical boundaries. In contrast, the EU's PilotR sandbox is a virtual sandbox covering the entire EU. The PilotR scheme, designed for market infrastructures based on DLT, does not provide a physical testing environment. Instead, it offers exemptions from EU financial law, allowing participating firms to test DLTs virtually across borders (Priem 2022; Zetzsche and Woxholth 2022).

The geographical reach of sandboxes can also be limited to the local or national level. Singapore's National Environment Agency ("NEA") has launched a sandbox for environmental services, allowing firms to test technologies aimed at reducing waste (NEA 2018). Within this sandbox, pilot trials have been conducted at various locations with different geographical limits, as determined by the NEA. One experiment was smaller-scale, involving an on-site waste gasification project at a nature park. In contrast, another trial within the same sandbox spanned the entire country, which involved collecting used cooking oil across Singapore (NEA 2018n.d.).

4.4. Monitoring

Regulators play a key role in monitoring sandbox participants. Monitoring allows regulators to understand the technologies involved in experiments and assess their effects. It also provides an opportunity to interact with participants to determine whether new regulations are needed to achieve objectives such as fostering innovation and ensuring consumer protection (Lemma 2020). In most cases, monitoring is essential to ensure that participants comply with sandbox regulations.

Monitoring has two main dimensions: frequency and intensity. Frequency refers to how often regulators interact with regulatees, and can range from weekly to every three months (Alaassar, Mention, and Aas 2020). Intensity refers to the level of detail and comprehensiveness involved in monitoring activities. This can range from less intensive measures, such as submitting information sheets and bi-monthly progress reports, to more intensive activities like scheduled face-to-face meetings between regulators and regulatees.

Over the course of a participant's involvement in a sandbox, monitoring may be dynamic, with frequency and intensity of monitoring varying over time. Less frequent and less intensive monitoring may suffice during low-risk stages, like planning experiments. During experimental phases, regulators may increase both frequency and intensity of monitoring to address potential risks. If regulators are unfamiliar with the technology, they may initially adopt higher frequency and intensity of monitoring to gain a better understanding, then reduce them once sufficient knowledge has been acquired. Monitoring, coupled with disclosure rules to increase transparency, will also be crucial to reduce the risk of riskwashing, whereby sandboxes are used as a stamp of marketability to reduce the perceived risk of innovations (Brown and Pirooska 2022; Chen and Taeihagh, *forthcoming*).

4.5. Exit rules

Exit rules are pre-determined conditions set by regulators that define when the sandbox experiment should be terminated. Firms violating these rules may be expelled from the sandbox and subject to penalties, such as fines or legal action. Examples of exit rules include discontinuing the experiment if the intended goals are not met, if the firm fails to comply with legal and regulatory requirements at the conclusion of the sandbox duration, or if the firm breaches agreed sandbox conditions (Gerlach and Rugilo 2019).

When regulators determine that an experiment fails to meet its objectives or does not perform as expected, they may either terminate it or allow it to continue. Continuing the experiment can offer regulators an opportunity to gain a deeper understanding of the innovation. In situations where the experiment is only marginally unsuccessful, regulators may allow firms to halt the experiment, amend the product or service, and resume testing, subject to regulatory approval.

Exit rules must consider consumer welfare and the risks experiments pose to the public. Regulators in Australia and Singapore have introduced exit rules allowing them to terminate experiments if they determine that the risks to consumers and the public outweigh the benefits (Didenko 2021; Gerlach and Rugilo 2019).

Consumers should have a clear channel for filing complaints if they experience detriment from sandboxes. This can be facilitated through a publicly accessible complaints mechanism, enabling consumers to notify regulators of any harm. Regulators should have the authority to terminate the experiment if it has been determined that the experiment has caused detriment (Poncibò and Zoboli 2020).

4.6. Transition plans

At the end of the sandbox duration, legal derogations expire, and firms must cease all sandbox activities. Regulators will then evaluate the product or service, provide feedback to the participant, and decide on a transition plan. Several options are available for the regulator from which to choose.

First, if the product or service meets safety standards, is reliable, provides public benefit, and is ready for real-world deployment, regulators may award a “go-to-market” decision. In this scenario, regulators must also determine whether new legislation is needed to ensure legal certainty for new products or services. If new or updated legislation is required, regulators can utilize the knowledge gained from the sandbox to draft legislation. Once the legislation is introduced, the firm may be granted a license to operate outside the sandbox. This decision is linked to confirmatory functionality, where meeting regulatory requirements leads to a “go-to-market” decision.

If the product or service is unsuitable for real-world deployment due to safety, reliability, or suitability concerns, regulators may reject all future applications for similar tests, effectively halting further experimentation. This suspension may remain until a new applicant proposes a pilot project with the potential for improved outcomes.

A final scenario is when the regulator determines that a product or service can be deployed with amendments and advises the firm on necessary changes. After modifying the product, the firm can re-apply to participate in a sandbox with confirmatory functionality. If successful, a “go-to-market” decision may be awarded.

Regulators should establish rules for recalling or retracting products or services. By the end of the sandbox period, they should have an improved understanding of new technologies’ risks and harms.

Regulatory limits should account for catastrophic or systemic failures and erratic product behavior. These limits may include environmental standards (e.g. emission limits for new engines), health and safety (e.g. accident rates for AVs), service reliability (e.g. maximum service outage frequency), minimum service standards (e.g. telemedicine call thresholds), digital security (e.g. fraud cases in digital payments), and personal data protection (e.g. limits on data privacy and processing).

If erratic behavior or system failure is identified, either through breaches of regulatory limits or unforeseen behavior, regulators can order firms to retract the product or service to mitigate potential harm. Following retraction, the product should undergo a separate investigation to determine the cause(s) of the issues. After identifying the source of the problems, regulators will decide whether

to terminate the product or allow rectification. Firms must propose amendments to address the identified issues and undergo confirmatory tests in a sandbox before relaunching the product or service in the real world.

When designing transition plans, regulators should consider the number of regulatory bodies involved. Products involving multiple sectors, such as blockchain innovations in healthcare, will require oversight from regulators in technology, data privacy, and healthcare to ensure comprehensive regulation during the transition.

As discussed in [Section 4.2](#) regarding multiple sandbox regulators, regulators involved in the entry and authorization stage should also participate in the transition plan. Firms meeting benchmarks set by all relevant regulators should be awarded a “go-to-market” decision.

Exceptions to transition plans may arise if a firm, despite regulatory approval for real-world deployment, decides not to proceed due to factors such as insufficient projected revenue, high operating costs, or a small market size. In such cases, the firm may choose to deploy the product or service elsewhere where conditions are more favorable. Regulators may also refrain from establishing a transition plan due to capacity and resource constraints.

4.7. Functionalities and characteristics

Exploratory functionality will require a longer duration for experimentation. The entry and authorization stage should have a greater degree of flexibility and may be collaborative. For instance, Singapore’s EMA sandbox allows amendments to applications after discussions with regulators (EMA 2019). A broad scope is required to support a wide range of experiments. This can vary from cross-sectoral initiatives like the European Blockchain Regulatory Sandbox which allows blockchain experiments across sectors such as healthcare and telecommunications, to more narrowly focused sandboxes like Singapore’s Ministry of Health’s Licensing Experimentation and Adaptation Program (LEAP), which targets healthcare innovations such as telemedicine or new healthcare models (Bhunia 2018; Mundell 2023). High frequency and intensity of monitoring can facilitate regulatory learning. Lower frequency and intensity are also possible: exploratory experiments with low public engagement (e.g. exploring AV technology in a closed circuit) may need less monitoring compared to experiments involving more consumers (e.g. FinTech experiment involving customers). Exit rules should allow for greater tolerance of nonperformance in meeting innovation benchmarks, which encourages experimentation. Experiments can be paused for adjustments, after which the amended experiment can be resubmitted for further trials. The availability or absence of transition plans does not significantly affect exploratory functionality. However, its absence may reduce the attractiveness of the sandbox to firms.

Confirmatory functionality may have a shorter duration and a narrower scope, focusing primarily on demonstrating an innovation’s capability to pass benchmarks. The design of entry and authorization criteria should focus on setting predefined benchmarks and assessing firms’ ability and appropriateness for sandbox entry. Hence, regulators involved should possess a deep understanding of the innovation and its risks before engaging in confirmatory tests. More frequent and intensive monitoring is required, as regulators must ensure that firms can deploy the innovation safely in the real world. Exit rules are stricter than exploratory functionality and should focus on predefined criteria that innovations must meet, rather than exploring use cases. Rules are stringent and based on firms’ ability to achieve specific standards. Regulators can introduce legislation to minimize regulatory delays and provide legal certainty, which facilitates seamless “go-to-market” decisions for transition plans.

On-demand functionality has a narrow scope and entry and authorization criteria, focusing on use cases or sectors of interest to the regulator. Duration depends on regulators’ needs and can be short or long term. There is greater emphasis on regulatory learning, which necessitates higher frequency and intensity of monitoring activities. Like exploratory functionality, moderate exit rules can encourage experimentation. Additionally, transition plans do not significantly affect on-demand functionality but can be included in the sandbox to attract firms.

5. Designing a regulatory sandbox

In this section, we propose a general framework for designing and implementing regulatory sandboxes (see Figure 2) and outline their life-cycle (see Figure 3). Design choices and functionalities can lead to varied positive or negative outcomes, with negative outcomes disproportionately affecting consumers, who far outnumber participating firms.

First, regulators should determine the sandbox's objective. Objectives can include exploring a new technology, validating a technology or product, addressing specific use cases, or refining regulations.

Next, regulators should decide if collaboration with other regulators is required. For example, general-purpose technologies such as exploring blockchain applications for peer-to-peer energy trading

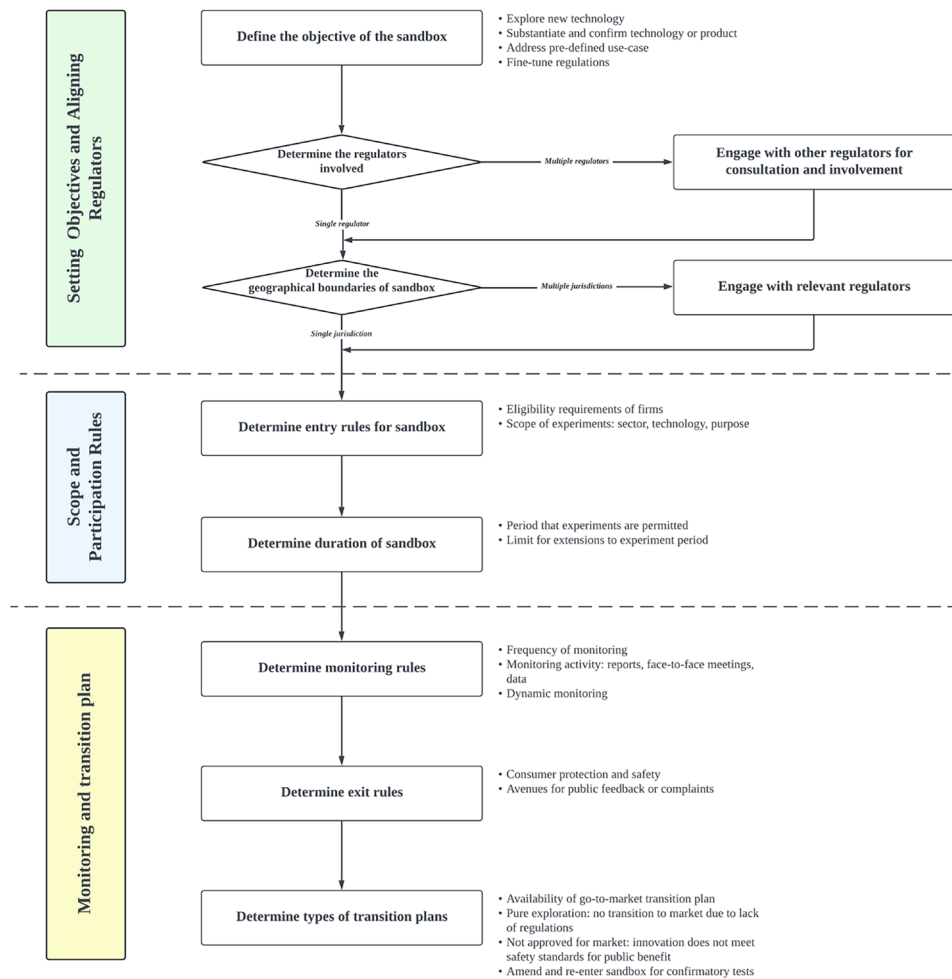


Figure 2. Process flow of designing and setting up a regulatory sandbox.

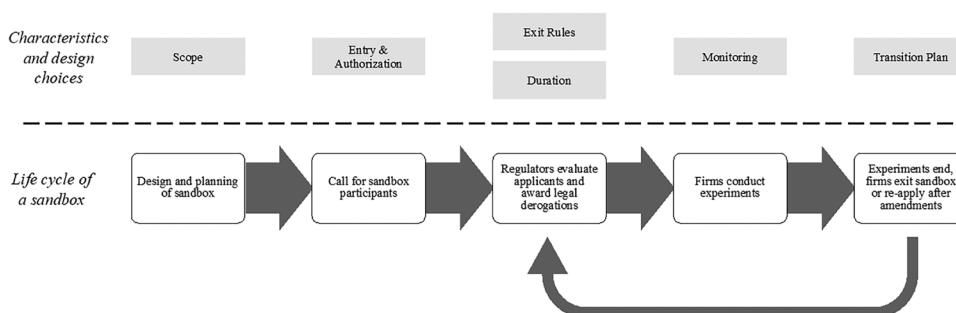


Figure 3. Life-cycle of a regulatory sandbox.

will require energy and technology regulators. Logistics-focused experiments for blockchain applications may involve port authorities or transport regulators.

Once relevant regulators are identified and consensus is reached, the sandbox's geographical or virtual boundaries should be defined. This may require coordinating with regulators from other jurisdictions. For example, federal agencies may need approvals from local authorities for experiments within physical boundaries.

The next step is to establish the entry rules, which include firm eligibility requirements and scope of experiments. Eligibility criteria can include firm characteristics (e.g. paid-up capital, ownership, location, workforce composition) and track record (e.g. relevant experience, compliance history, criminal records of directors). Regulators should also define the sandbox scope, with limits on the technology, sector, industry, and objectives of experiments. Regulators should also be aware that sandboxes with lenient entry rules, such as automatic approvals, pose higher risks due to insufficient screening mechanisms. Automatic approvals may admit firms lacking technical expertise, which increases the likelihood of failed experiments.

Regulators should consider the duration of sandbox experiments. Key considerations are the period of experiments and whether extensions are permitted. If extensions are allowed, regulators must decide on the mechanism: whether automatic approval is granted upon request or a formal review process is introduced.

Monitoring rules should align with regulators' needs. Achieving greater regulatory learning will require higher monitoring intensity and firm engagement. High-risk experiments will require stricter monitoring, with higher frequency and greater scrutiny. Low monitoring intensity and inadequate exit rules (penalties) increase public risk, as firms may act opportunistically and conduct riskier experiments without fear of consequences.

The sandbox should include exit rules to safeguard the public and ensure experiments generate public benefit. For experiments involving the public, such as FinTech investment products, additional mechanisms like a public feedback or complaint portal will be necessary to protect the public interest.

Finally, regulators should determine transition plans available within the sandbox, which are related to confirmatory functionality. For sandboxes with high exploratory functionality but low confirmatory functionality (e.g. sandboxes for exploring with few confirmatory tests due to uncertainty over innovation), regulators can exclude go-to-market transition plans. Regulators can instead provide an option to amend experiments and reenter the sandbox, subject to regulatory approval.

Regulators should recognize that poorly designed sandboxes can negatively affect private firms. Lax entry rules and insufficient monitoring may allow firms to misuse the sandbox status for fraudulent fundraising activities, or to remain inactive without genuine innovation. This can result in low-quality experiments, a lack of public benefit, and regulatory costs outweighing the value generated by the sandbox.

Some firms may join sandboxes to gain advantages over nonparticipating firms, such as access to grant funding. For example, IMDA's PET sandbox offers up to 50% cost coverage for pilot projects. However, its 6-month no-extension limit may deter firms from participating in the sandbox solely for favorable treatment (IMDA 2022).

The design phase represents the first stage in the sandbox's life-cycle. Once the design is ready, regulators announce a call for participants and evaluate applicants, either on a rolling basis or by cohorts. Successful applicants receive legal derogations and begin experiments, while being subjected to monitoring requirements and pre-determined exit rules. At the end of the experimentation period, firms must exit the sandbox, with three outcomes possible: termination, re-apply after amendments to experiments, or a go-to-market approval. Firms seeking further experiments should be able to apply to reenter the sandbox.

6. Conclusion

The article presents a general framework for designing regulatory sandboxes. First, a decision flow process for introducing sandboxes is presented, based on considerations over the grounds for a legal experiment, the availability of experimental legislation, and the appropriateness of existing regulations.

The next step involves determining what functionalities the sandbox should have based on regulators' objectives and goals. Three types of functionalities are explored: exploratory, confirmatory, and on-demand. These functionalities can be combined to achieve a greater range of objectives.

We suggest that functionalities emerge because of distinct design choices and calibrations of sandbox characteristics. To our knowledge, this is the first paper that presents a general sector-agnostic framework for designing sandboxes, which integrates functionalities and characteristics. Our findings can be valuable to regulators in understanding and designing sandboxes. However, our study has its limitations. Notably, the capacity and ability of regulators in designing and implementing sandboxes may negatively affect sandbox design and outcomes. Future research could examine the relationship between capacity and design, the potential of sandboxes as a new policy instrument, and the relationship between sandbox design and outcomes.

CRediT authorship contribution statement

Gabriel Kwok Hui Chen: Methodology, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing. Araz Taeihagh: Conceptualization, Methodology, Investigation, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & Editing, Funding acquisition.

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Appendix A

Table A1. Sandboxes, jurisdictions, sectors, and their primary functionality.

Sandbox	Jurisdiction	Sector	Primary Functionalities	Reference
Autonomous Vehicle Sandbox	Singapore	Transport	Exploratory Confirmatory	Tan and Taeiagh 2021a
European Blockchain Sandbox for Distributed Ledger Technologies	European Union	Distributed Ledger Technology	On-demand	European Commission 2023
PilotR Sandbox	European Union	Distributed Ledger Technology	Exploratory	Priem 2022; Zetzsche and Woxholth 2022
Experiments Decentralized, Sustainable Electricity Production (EDSEP)	Netherlands	Energy	Exploratory	Correa, Gomez, and Cossent 2021; van der Waal, Das, and van der Schoor 2020
Energy Market Authority Regulatory Sandbox	Singapore	Energy	Exploratory	EMA 2019
Energy Market Authority Thematic Sandbox	Singapore	Energy	On-demand	EMA 2019
National Environment Agency Sandbox	Singapore	Environment	Exploratory	NEA 2018
Enhanced Regulatory Sandbox	Australia	FinTech	Exploratory	Alaassar, Mention, and Aas 2021; Didenko 2021
FinTech Supervisory Sandbox	Hong Kong	FinTech	Exploratory	Alaassar, Mention, and Aas 2021
MNB FinTech Regulatory Sandbox	Hungary	FinTech	Exploratory	Kálmán 2025
Reserve Bank of India Sandbox	India	FinTech	Exploratory	Goo and Heo 2020; Ahern 2021
Financial Technology Regulatory Sandbox	Malaysia	FinTech	Exploratory	Goo and Heo 2020
Modelos Novedosos	Mexico	FinTech	Confirmatory	Larios-Hernandez, 2025
Monetary Authority of Singapore Sandbox	Singapore	FinTech	Exploratory	Goo and Heo 2020; Kálmán 2025
Bank of Thailand Regulatory Sandbox	Thailand	FinTech	Exploratory	Goo and Heo 2020
Financial Conduct Authority Sandbox	United Kingdom	FinTech	Exploratory	Gerlach and Rugilo 2019; Tsai, Lin, and Liu 2020; Kálmán 2025
Sandbox Korea Financial Regulatory Sandbox	South Korea	Fintech, ICT, and Industrial convergence	Exploratory	Lee and Seo 2022
Licensing Experimentation and Adaptation Program	Singapore	Healthcare	Exploratory	Bhunja 2018; Mundell 2023
Privacy Enhancing Technologies Sandbox	Singapore	ICT	On-demand	IMDA 2022
InsurTech Sandbox	Hong Kong	Insurance	Exploratory	Ahern 2021